



DEFIHUB

OPTION A DEFI SUPER-APP

Blockchain Technologies 2 —
Final Project

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PROBLEM & GOAL

Problem:

DeFi is fragmented across separate protocols:

- DEX for swaps
- Lending platforms
- Yield vaults
- Governance systems

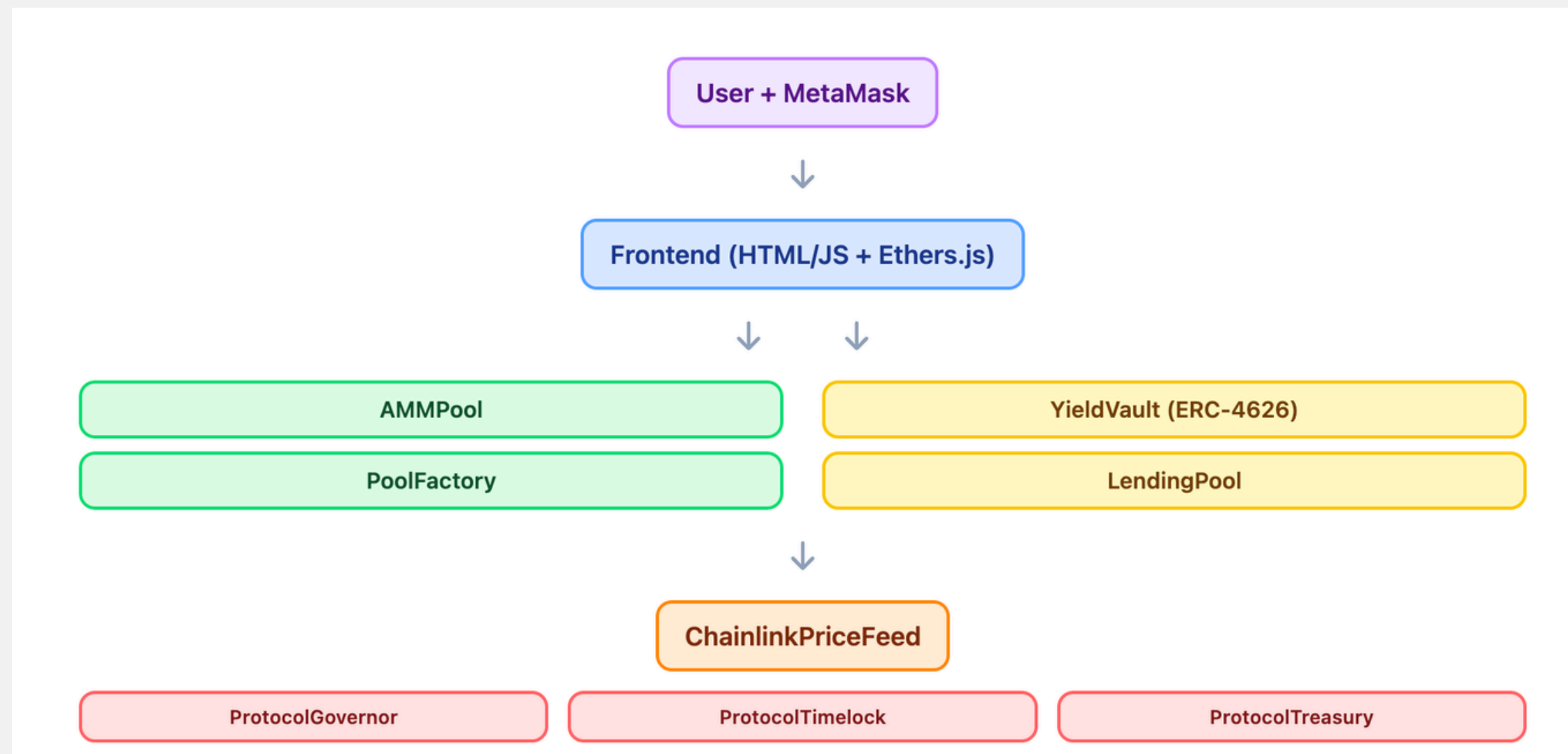
Our Solution:

Unified DeFi Ecosystem

- Swap, borrow, yield, DAO in one stack
- Modular smart contracts
- Layer 2 deployment (Arbitrum)
- The Graph indexing
- Integrated dApp frontend

User Flow: MetaMask → Frontend → Arbitrum Sepolia Contracts → The Graph Indexing

ARCHITECTURE OVERVIEW



AMM: HOW SWAP WORKS

Mechanics

Formula: $x \cdot y = k$

Fee: 0.3% (997/1000)

LP Tokens: Minted on add, burned on remove

Protection:

- Slippage control(amountOutMin)
- Reentrancy guard
- CEI pattern on with drawals

User Flow

1. Approve tokenA → Pool

2. `swapExactTokensForTokens(tokenIn, amountIn, amountOutMin, deadline)`

3. Pool updates reserves

4. Receive tokenOut

Pool: 0x27a1...8719

mUSDC: 0xEc0C...

mWETH: 0x05aa...



LENDING: LTV, HEALTH FACTOR, LIQUIDATION

Parameters	
Parameter	Value
Max LTV	75%
Liquidation threshold	80%
Borrow rate	10% APR
Liquidation bonus	5%
Collateral	mWETH
Borrow asset	mUSDC
Oracle	Chainlink

HOW IT WORKS

Security: Staleness check prevents borrowing at outdated prices (audit finding S-03)

- 1. depositCollateral**
Deposit WETH into pool
- 2. borrow**
Get USDC if debt ≤ LTV by oracle price
- 3. Health factor**
Monitors collateral value vs debt
- 4. liquidate**
Liquidator repays debt, gets collateral + 5% bonus



YIELD VAULT (ERC-4626)

Standard ERC-4626

Protocol: Tokenized Vault

Flow: deposit(assets) → shares

Underlying: mUSDC

Owner: Timelock

Security: _decimalsOffset = 3

Protection from inflation attack



User Flow

Step 1
Approve USDC → Vault

Step 2
deposit(amount, receiver)

Step 3
Balance of shares grows

Step 4
withdraw/redeem assets

Contract:
0x280E3BcF84299D8f53eB7876a4daA2bAe9d40124



ORACLE (CHAINLINK)

ChainlinkPriceFeed Adapter

Security Checks

- ✓ answer > 0
- ✓ Full round completed
- ✓ Staleness check ≤ 1 day

Network:
Arbitrum Sepolia testnet
If no env feed → MockV3Aggregator

Adapter: `0x0e7Db0BD600bB8A3919302410F55ACA81B15632B`

Why Not AMM Price?

Source	Flash Loan	Staleness
AMM Spot	Yes ✗	—
Chainlink	No ✓	Blocked ✓

Token	DFH (ERC20Votes)
Quorum	4%
Proposal threshold	1% supply
Voting delay	~1 day (7200 blocks)
Voting period	~1 week (50400)
Timelock delay	2 days

1. Create

Holder with $\geq 1\%$ tokens

2. Delegate

Snapshot votes (anti flash-loan)

3. Vote

castVote during period

4. Queue

Success $\rightarrow$ queue in Timelock

GOVERNANCE (DAO)



FACTORY & UUPS UPGRADE

PoolFactory

Address: 0x7612...

createPool
Standard CREATE deployment

createPoolDeterministic
CREATE2 with salt

predictDeterministicAddress
Calculate address before deploy

Accelerates deployment of new pairs

UUPS Config

Proxy: 0x3907...

Protocol Parameters
feeBps, protocolName, etc.

Upgrade Path
Only onlyOwner (Timelock)

Pattern
Implementation + ERC1967Proxy

Upgrade path without full migration

DEPLOY PIPELINE (L2)

```
$ ./scripts/deploy-arbitrum-sepolia.sh  
# → forge script Deploy.s.sol --broadcast --verify  
# → deployments/arbitrum-sepolia.json
```

Deployment Order:

1. Mock tokens + GovernanceToken

2. Timelock + Governor + Treasury

3. Oracle (Chainlink or Mock)

4. Factory + Pool + liquidity

5. Vault + LendingPool

6. UUPS config proxy

7. Treasury registerModule

8. Ownership → Timelock

16+ Verified Contracts

on Arbitrum Sepolia Explorer

FRONTEND DAPP

FEATURES

1. Connect Wallet

MetaMask connection

2. Switch Network

Auto-switch to Arbitrum Sepolia

3. Swap Tokens

mUSDC ↔ mWETH

4. Vault Deposit

Earn yield on deposits

5. Governance

Delegate & Vote on proposals

TECH STACK

- HTML/JS + Ethers.js
- MetaMask integration
- Arbitrum Sepolia (0x66eee)

TESTING & CI

Metrics

113

Total Tests

unit + fuzz + invariant + fork

97.89%

Line Coverage

GitHub Actions

CI Pipeline

build, test, coverage, fmt

Test Types

Unit Tests

AMM math, vault shares, lending LTV

Fuzz Tests

Swap amounts, deposit bounds

Invariant Tests

Lending solvency, pool constant k

Fork Tests

L2 behavior, Chainlink integration

SECURITY & DELIVERABLES

Audit Findings

ID	Finding	Severity	Status
S-01	Reentrancy removeLiquidity	High	Fixed ✓
S-02	Unprotected UUPS initializer	High	Fixed ✓
S-03	Chainlink staleness check	Medium	Fixed ✓
S-04	Factory zero-address check	Low	Fixed ✓
S-05	Missing config event	Info	Fixed ✓

✓ Completed

- ✓ Smart contracts + tests
- ✓ L2 deploy + verified addresses
- ✓ Frontend dApp
- ✓ Subgraph config
- ✓ README documentation

📄 Documentation

- 📄 Audit report (8+ pages)
- 📄 Architecture doc (6+ pages)
- 📄 Gas optimization report
- 📄 This presentation (PDF)

Demo & Questions?

Live Demo

- ✓ MetaMask Connection
- ✓ Token Swap
- ✓ Vault Deposit
- ✓ Arbiscan Tx Links

Repository

github.com/alseytdastan/final_blockchain_technologies

Links

Frontend (GitHub Pages)
Pool on Arbiscan
README deploy table

Thank You!

Questions & Discussion